

Beyond Osimertinib Alone: CNS-Only Relapse Despite Negative tDNA Surveillance in Resected *EGFR*-Mutant Lung Adenocarcinoma

Shreya Patel, DO¹ | Angie Confesor, MS^{3,2} | Vasu Malhotra, DO¹ | Imran Ahmad, MD¹

¹Lakeland Regional Medical Center, Lakeland, FL | ²Orlando College of Osteopathic Medicine, Orlando, FL

BACKGROUND

- Adjuvant osimertinib is standard of care for resected *EGFR*-mutant NSCLC (ADAURA trial).
- ctDNA is increasingly used for post-surgical surveillance, yet its sensitivity in CNS-only disease is uncertain.
- Emerging data (FLAURA2, MARIPOSA) support intensified upfront strategies in high-risk *EGFR*-mutant advanced disease.
- We present a patient with early-stage *EGFR*-mutant lung adenocarcinoma who developed isolated hemorrhagic brain metastases despite repeatedly negative ctDNA while on adjuvant osimertinib.



This case underscores the need to refine risk stratification, CNS surveillance, and the integration of intensified therapy beyond osimertinib alone

CASE SUMMARY TIMELINE



Presentation: Cough, sore throat, dyspnea on exertion. CT chest: 2 cm LLL nodule near hilum, PET SUV 18. No nodal disease.

Robotic LLL lobectomy + lingulectomy (margins close). **Pathology:** Poorly differentiated adenocarcinoma, acinar predominant with mucinous, papillary, micropapillary, cribriform components. Stage IA3 (pT1cN0M0), +STAS.

Adjuvant chemotherapy: Carboplatin + Pemetrexed (Completed 3 of 4 planned cycles due to tolerability).

Adjuvant osimertinib initiated.

Serial ctDNA (Guardant360 & Reveal): No tumor-related somatic alterations detected. CT surveillance: No evidence of disease. Persistent headaches.

ctDNA (08/28/24): Not detected, TF 0%.

Brain MRI (08/30/24): Multiple supratentorial enhancing lesions (concerning for hemorrhagic metastases).

PET/CT (09/11/24): No extracranial disease.

Gamma Knife stereotactic radiosurgery to brain lesions.

Systemic therapy escalated to Amivantamab + Lazertinib.

Ongoing surveillance with imaging and ctDNA. Stable disease as of 09/2025.

MOLECULAR & PATHOLOGY PROFILE

MOLECULAR / BIOMARKER PROFILE

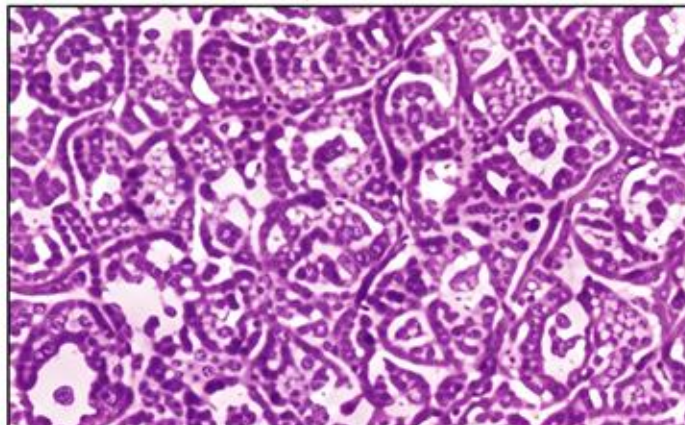
- EGFR E746_A750del** (Exon 19 deletion) Detected by Guardant360 (0.2%) pre-operatively
- PD-L1 TPS** <1% (No expression)
- CARIS NGS:** *EGFR* Exon 19 del, *TP53*, *RB1*

PATHOLOGIC HIGH-RISK FEATURES

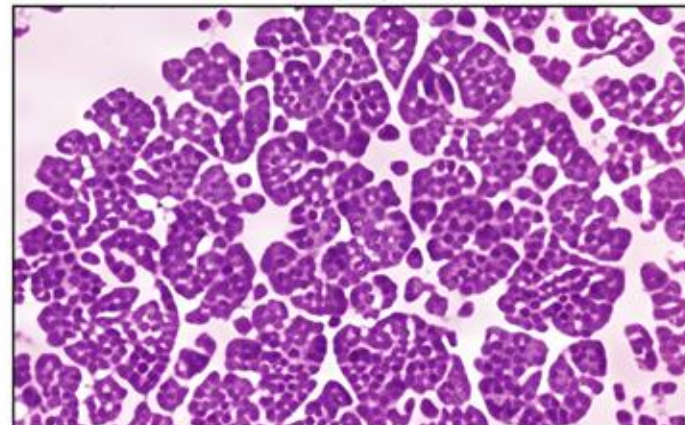
- Poorly differentiated adenocarcinoma
- Micropapillary component
- Spread through air spaces (STAS)+
- Close margins requiring lingulectomy
- Co-alterations: *TP53*, *RB1*
- ctDNA detectable despite minimal disease

★ Multidisciplinary Tumor Board Recommendation: Adjuvant chemotherapy followed by osimertinib given high-risk pathologic and molecular features.

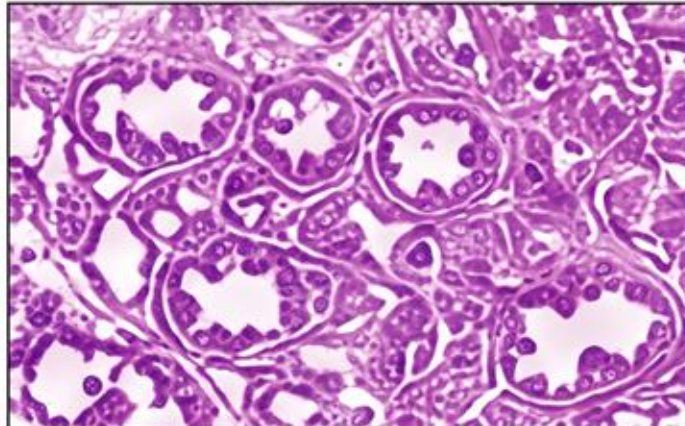
Acinar predominant



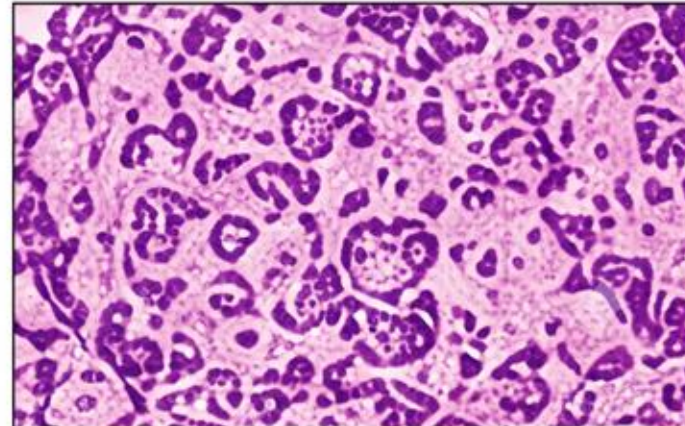
Micropapillary



Mucinous features



Cribriform pattern



PATIENT SNAPSHOT



62-year-old woman, never-smoker

Physician by profession

PMHx: Asthma, diverticulitis, tinnitus

Family Hx: Gastric cancer (grandmother)

12/2022: 2 cm LLL nodule (SUV 18), no nodal disease

KEY POINT: DISCORDANCE

Despite repeatedly negative ctDNA and no extracranial disease, the patient developed multifocal CNS metastases.

Likely explanation: Limited ctDNA shedding from CNS disease across the blood-brain barrier.

POTENTIAL EXPLANATIONS FOR DISCORDANCE



Blood-brain barrier may limit ctDNA release from intracranial lesions.



Low tumor burden within CNS may be below detection threshold.



Delayed ctDNA positivity after radiosurgery may reflect treatment-induced tumor cell lysis and BBB disruption.

IMAGING TIMELINE (KEY STUDIES)

	12/2022 (Diagnosis)	08/30/2024 (CNS Relapse)	01/29/2025 (Post Treatment)
CT Chest			
PET/CT			
Brain MRI	Not Performed		

No extracranial disease on serial PET/CT. Multiple hemorrhagic intracranial metastases detected only on MRI.

TREATMENT COURSE & OUTCOME

Therapy	Dates	Details / Outcome
Adjuvant Chemotherapy	01/2023 – 03/2023	Carboplatin + Pemetrexed. Completed 3 of 4 cycles. (Tolerability concerns)
Adjuvant Osimertinib	04/2023 – 08/2024	Well tolerated. ctDNA undetectable. No radiographic disease.
Gamma Knife Radiosurgery	09/28/2024	To multiple brain metastases.
Amivantamab + Lazertinib	09/2024 – Present	Initiated due to CNS progression on osimertinib. Amivantamab dose reduced for Grade 3 dermatologic toxicities; intermittently held (02–05/2025) for vulvar granulomatous lesion; then restarted at lowest effective dose.
Current Status	09/2025	PET: No evidence of disease. Brain MRI: Stable post-treatment changes. Guardant Reveal: No ctDNA detected. Guardant360 (08/11/2025): TP53 R249S detected; no <i>EGFR</i> alteration. PET (09/03/2025): Stable, no progression; small micronodules.

KEY DISCUSSION POINTS

- Does ctDNA positivity in early-stage NSCLC redefine "high-risk" disease?
- Which patients are best candidates for intensified monitoring and early therapeutic escalation based on ctDNA?
- What is the optimal role of liquid biopsy in post-surgical surveillance of *EGFR*-mutant NSCLC?
- Should baseline brain MRI be considered even in early-stage *EGFR*-mutant NSCLC with high-risk features?
- How should we interpret discordance between ctDNA negativity and radiographic CNS-only progression?
- Does isolated brain metastasis indicate need for immediate escalation beyond osimertinib?

TAKE-HOME MESSAGES



ctDNA may not reliably detect CNS-only metastases due to the blood-brain barrier.



High-risk pathologic and molecular features in early-stage *EGFR*-mutant NSCLC warrant closer surveillance, including consideration of baseline brain MRI.



Discordance between liquid biopsy and imaging is a real-world challenge requiring clinical judgment and multimodality assessment.



Frontline and adjuvant intensification strategies, including amivantamab + lazertinib, may be appropriate for biologically aggressive disease.

CONCLUSION

This case highlights a patient with resected *EGFR* exon 19–mutant lung adenocarcinoma who experienced isolated hemorrhagic CNS relapse despite negative ctDNA surveillance while on adjuvant osimertinib. It underscores the limitations of ctDNA in CNS disease detection, the prognostic significance of adverse molecular/pathologic features even in stage I disease, and the importance of individualized surveillance and therapeutic strategies. Ongoing studies are needed to define optimal integration of ctDNA, imaging, and intensified therapy in *EGFR*-mutant NSCLC.



Abbreviations: NSCLC: non-small cell lung cancer; EGFR: epidermal growth factor receptor; ctDNA: circulating tumor DNA; STAS: spread through air spaces; BBB: blood-brain barrier; SRS: stereotactic radiosurgery.

References:
(FLAURA2).

Herbst RS, et al. N Engl J Med. 2020;383:1711-1723 (ADAURA).

Janne PA, et al. N Engl J Med. 2024;291:250-263 (MARIPOSA).

Zhou C, et al. N Engl J Med. 2023;389:2039-2051

Remon J, et al. Ann Oncol. 2023;34:339-351.

FMA 2025